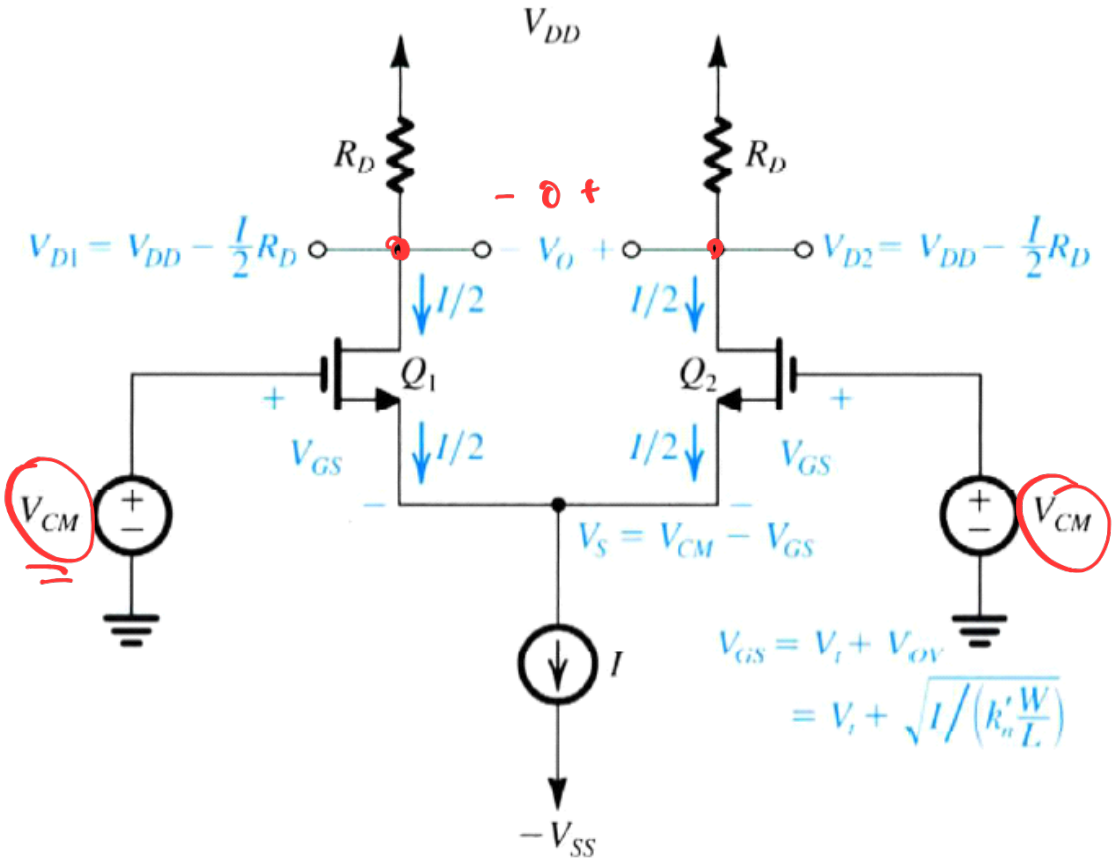
Reading

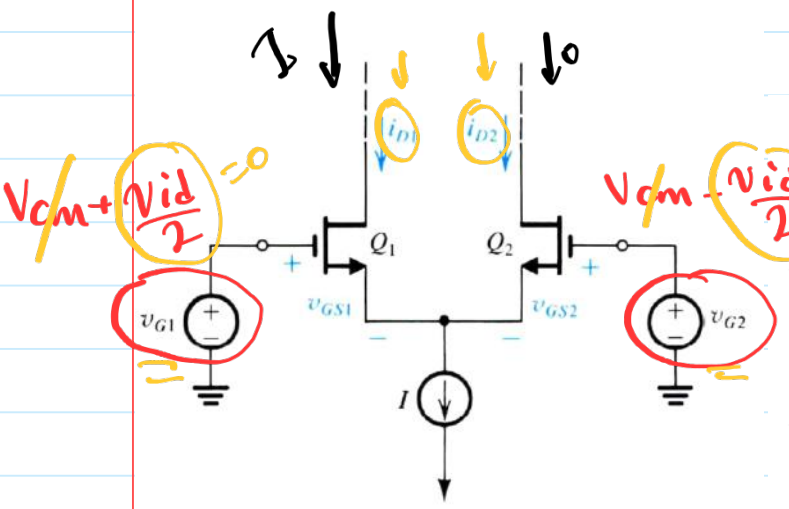
Adel S. Sedra and Kenneth C. (KC) Smith, Tony Chan Carusone, and Vincent Gaudet
Microelectronic Circuits 8th Edition, *Oxford University Press*, 2019.

- Chapter 9.1 The MOS Differential Pair
- Chapter 9.2 The BJT Differential Pair

Recap


V_{DD}
R_D
R_D
$V_{D1} = V_{DD} - \frac{I}{2}R_D$
V_O
$V_{D2} = V_{DD} - \frac{I}{2}R_D$
$I/2$
$I/2$
Q_1
Q_2
V_{GS}
$I/2$
$I/2$
V_{GS}
V_{CM}
V_{CM}
$V_S = V_{CM} - V_{GS}$
I
$V_{GS} = V_t + V_{OV}$
$= V_t + \sqrt{I/\left(k_n' \frac{W}{L}\right)}$
$-V_{SS}$

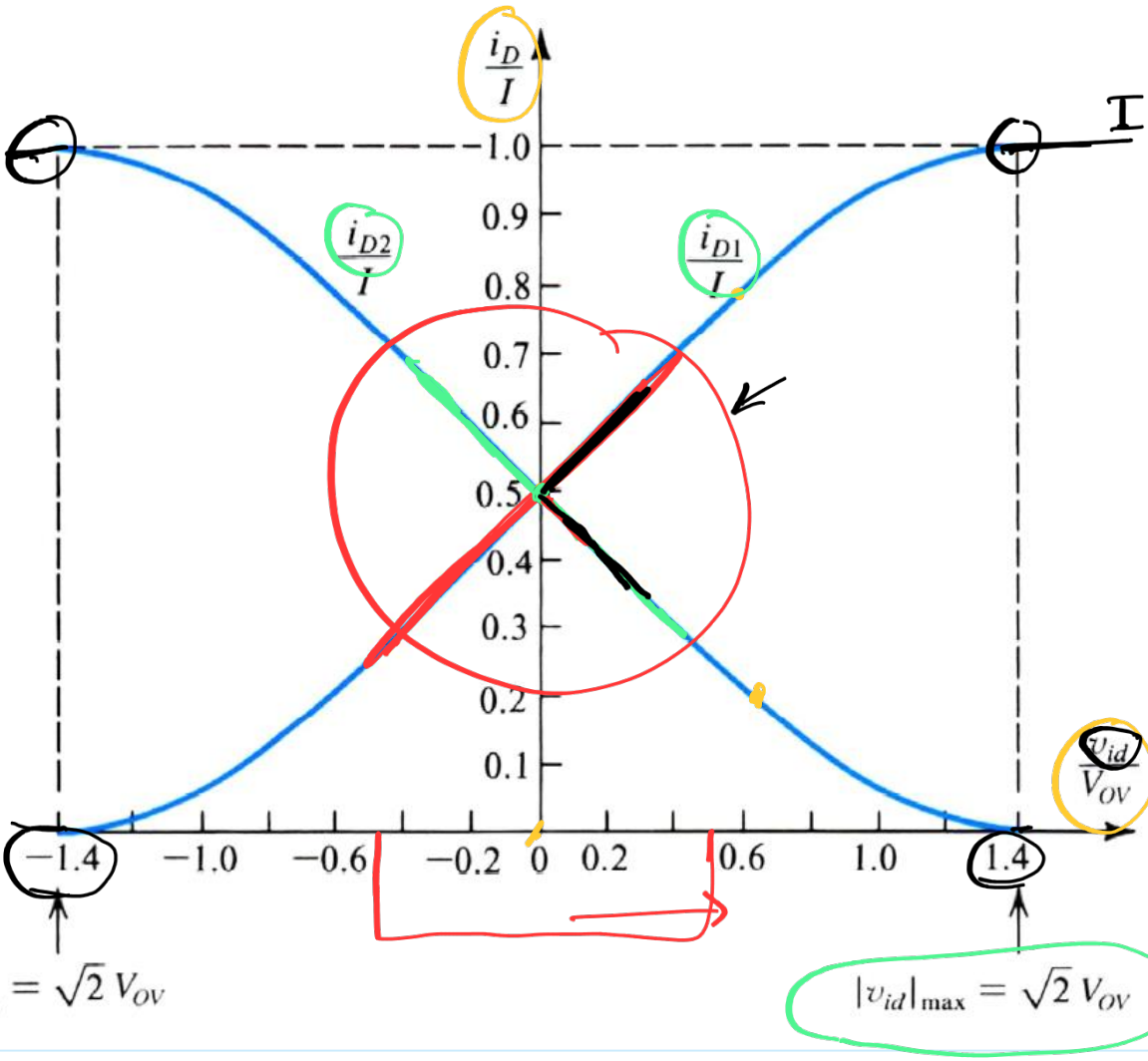
Recap



$$v_{GS1} - v_{GS2} = v_{G1} - v_{G2} = v_{id}$$

$$i_{D1} = \frac{I}{2} + \left(\frac{I}{V_{OV}}\right)\left(\frac{v_{id}}{2}\right)\sqrt{1 - \left(\frac{v_{id}/2}{V_{OV}}\right)^2}$$

$$i_{D2} = \frac{I}{2} - \left(\frac{I}{V_{OV}}\right)\left(\frac{v_{id}}{2}\right)\sqrt{1 - \left(\frac{v_{id}/2}{V_{OV}}\right)^2}$$



$|v_{id}|_{max} = \sqrt{2}\,V_{OV}$

$|v_{id}|_{max} = \sqrt{2}\,V_{OV}$

Recap

$$i_{D1} = \frac{I}{2} + \left(\frac{I}{V_{OV}}\right)\left(\frac{v_{id}}{2}\right)\sqrt{1 - \left(\frac{v_{id}/2}{V_{OV}}\right)^2}$$

$$i_{D2} = \frac{I}{2} - \left(\frac{I}{V_{OV}}\right)\left(\frac{v_{id}}{2}\right)\sqrt{1 - \left(\frac{v_{id}/2}{V_{OV}}\right)^2}$$

$v_{id} \ll 2V_{OV}$

$$i_{D1} \simeq \frac{I}{2} + \left(\frac{I}{V_{OV}}\right)\left(\frac{v_{id}}{2}\right)$$

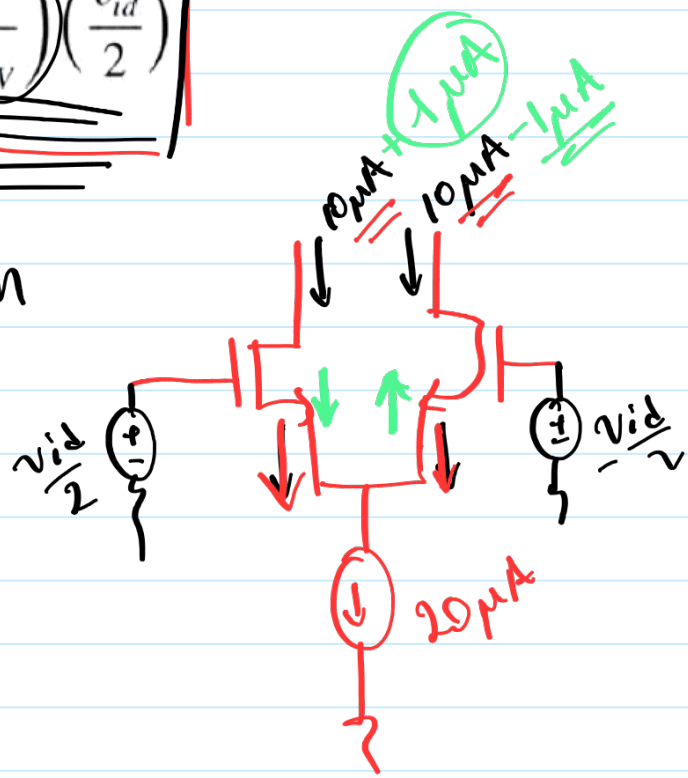
$$i_{D2} \simeq \frac{I}{2} - \left(\frac{I}{V_{OV}}\right)\left(\frac{v_{id}}{2}\right)$$

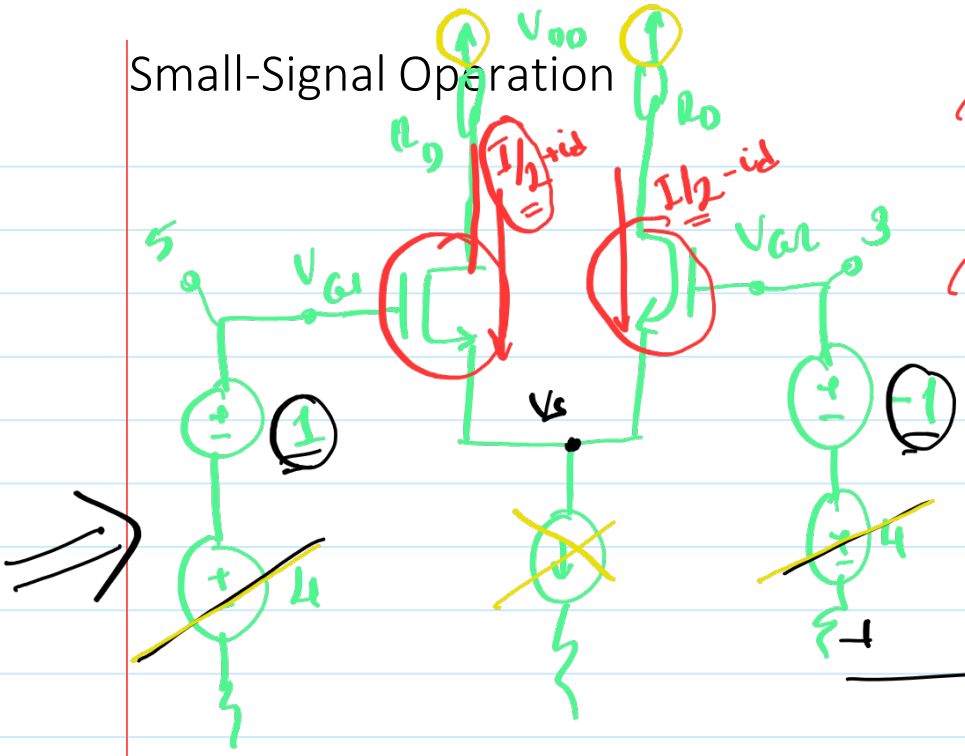
$$i_d = \left(\frac{I}{V_{OV}}\right)\left(\frac{v_{id}}{2}\right)$$

G_m

$i_{D1} = \frac{I}{2} + i_d$

$i_{D2} = \frac{I}{2} - i_d$





$$V_{G1} = V_{cm} + \frac{1}{2} v_{id}$$

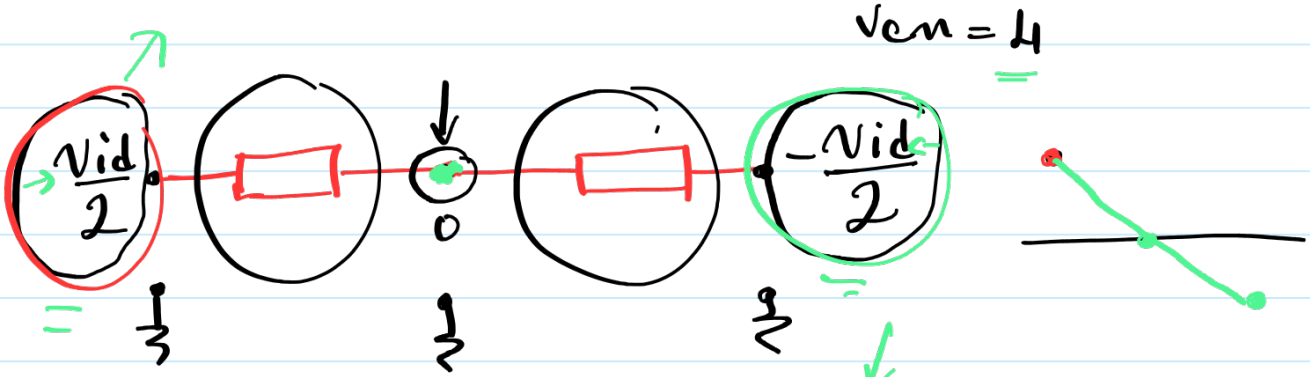
$$V_{G2} = V_{cm} - \frac{1}{2} v_{id}$$

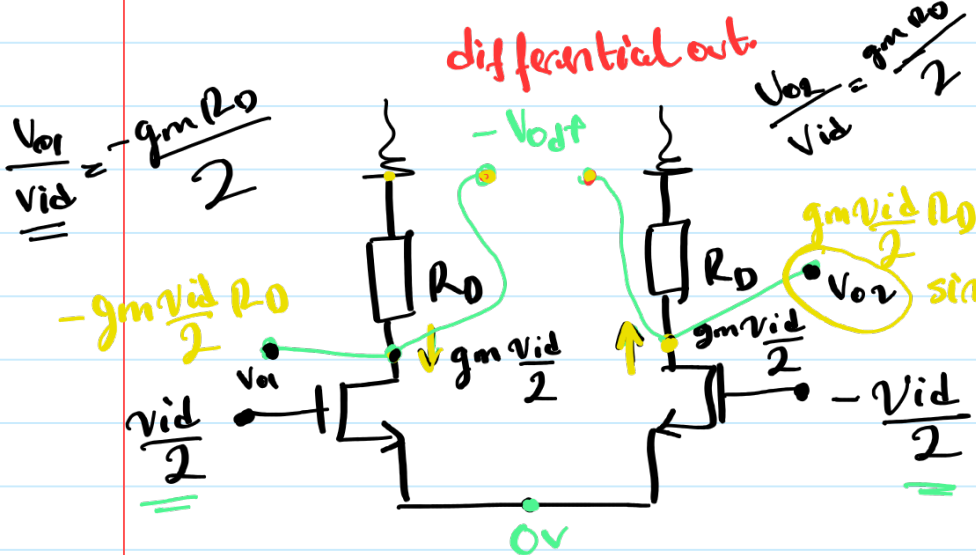
$$5 = V_{cm} + \frac{1}{2} v_{id} \quad (4)$$

$$3 = V_{cm} - \frac{1}{2} v_{id} \quad (2)$$

$$8 = 2V_{cm} + 0$$

$$V_{cm} = 4$$





$$\frac{V_{o1}}{v_{id}} = -\frac{g_m R_D}{2}$$

$$\frac{V_{o2}}{v_{id}} = \frac{g_m R_D}{2}$$

$$-\frac{g_m v_{id} R_D}{2}$$

$$i_d = \left(\frac{I}{V_{ov}}\right) \frac{v_{id}}{2}$$

$$g_m = \frac{2I_D}{V_{ov}} = \frac{2I/2}{V_{ov}}$$

$$g_m = \frac{I}{V_{ov}}$$

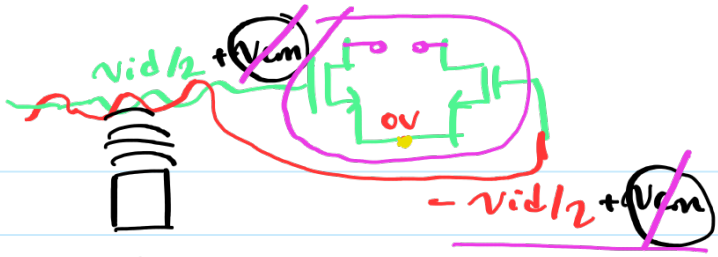
$$V_{od} = g_m \frac{v_{id}}{2} R_D + g_m \frac{v_{id}}{2} R_D$$

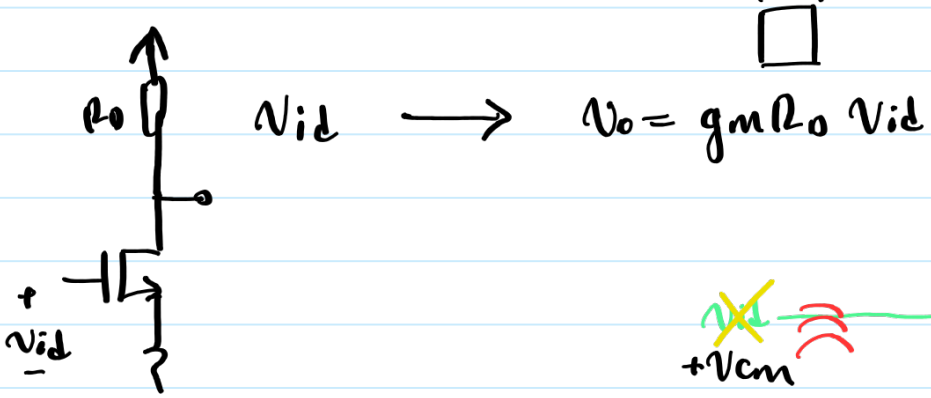
$$i_d = g_m \left(\frac{v_{id}}{2}\right)$$

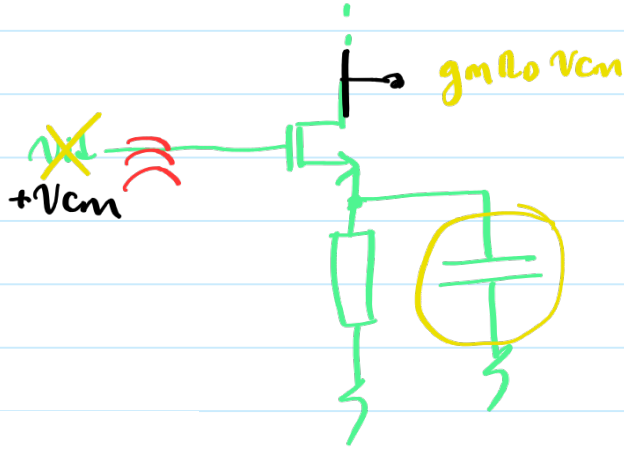
$$V_{od} = g_m v_{id} R_D$$

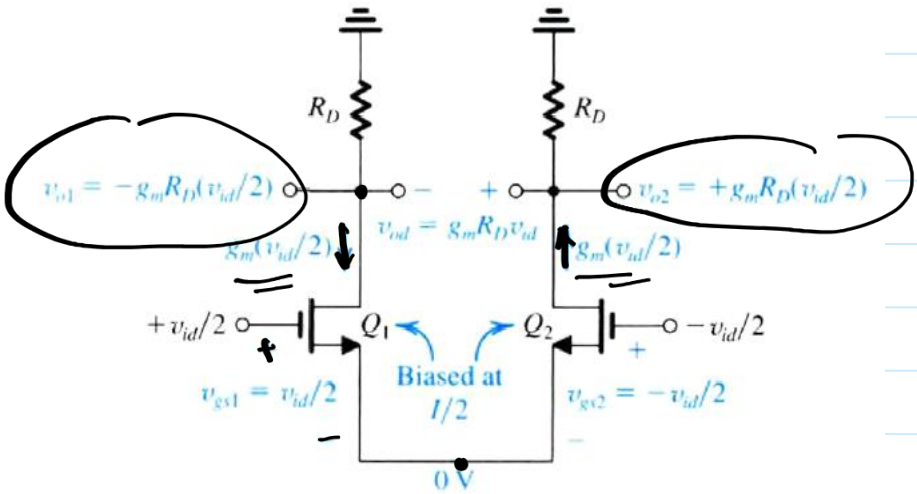
$$A_d = \frac{V_{od}}{v_{id}} = \frac{g_m v_{id} R_D}{v_{id}} = g_m R_D$$

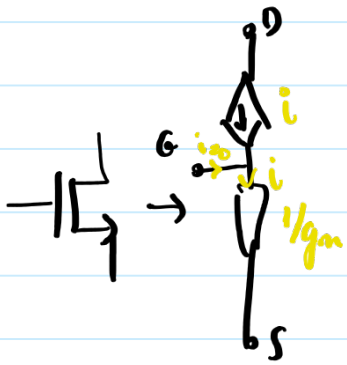
Small-Signal Operation

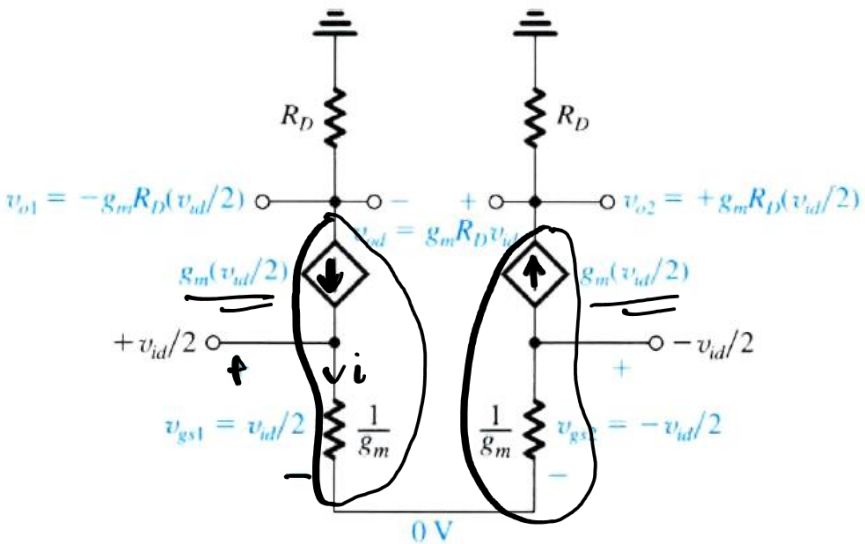




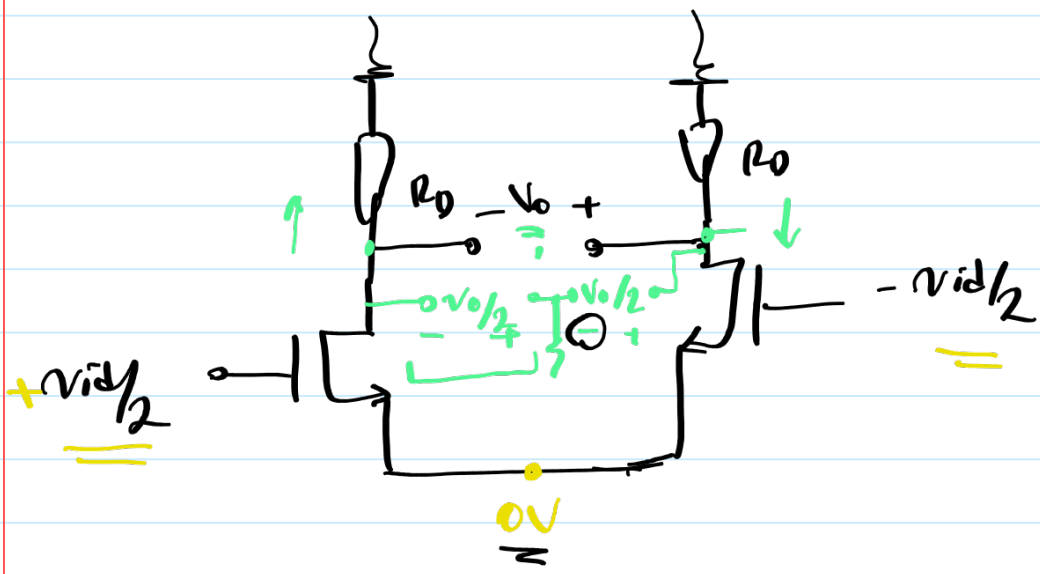


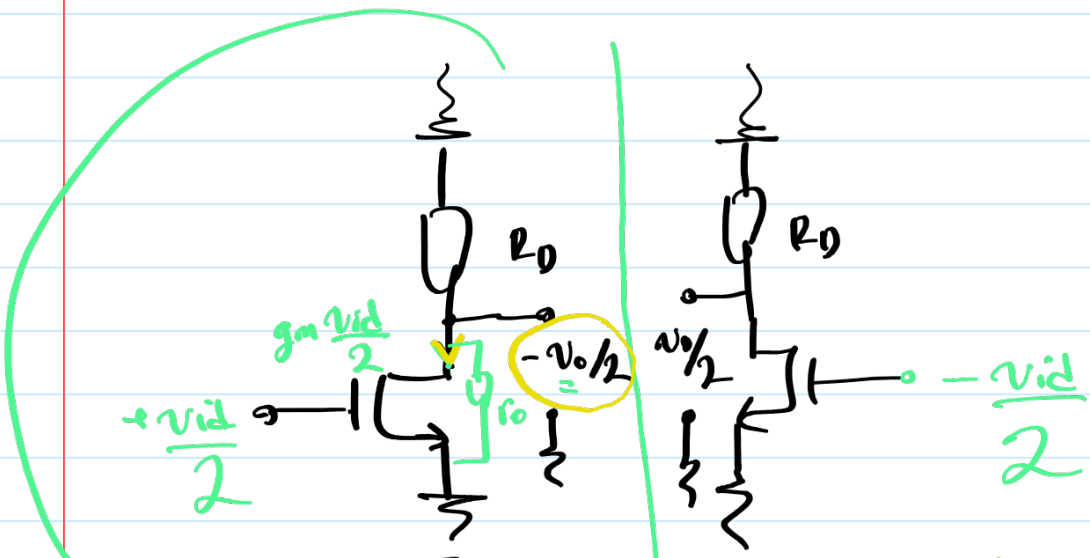






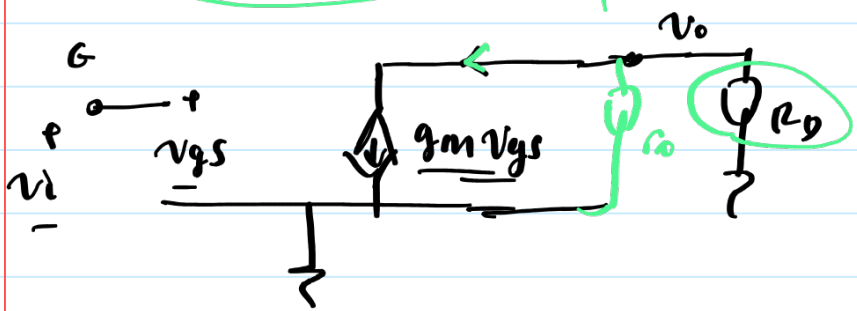
The Differential Half Circuit



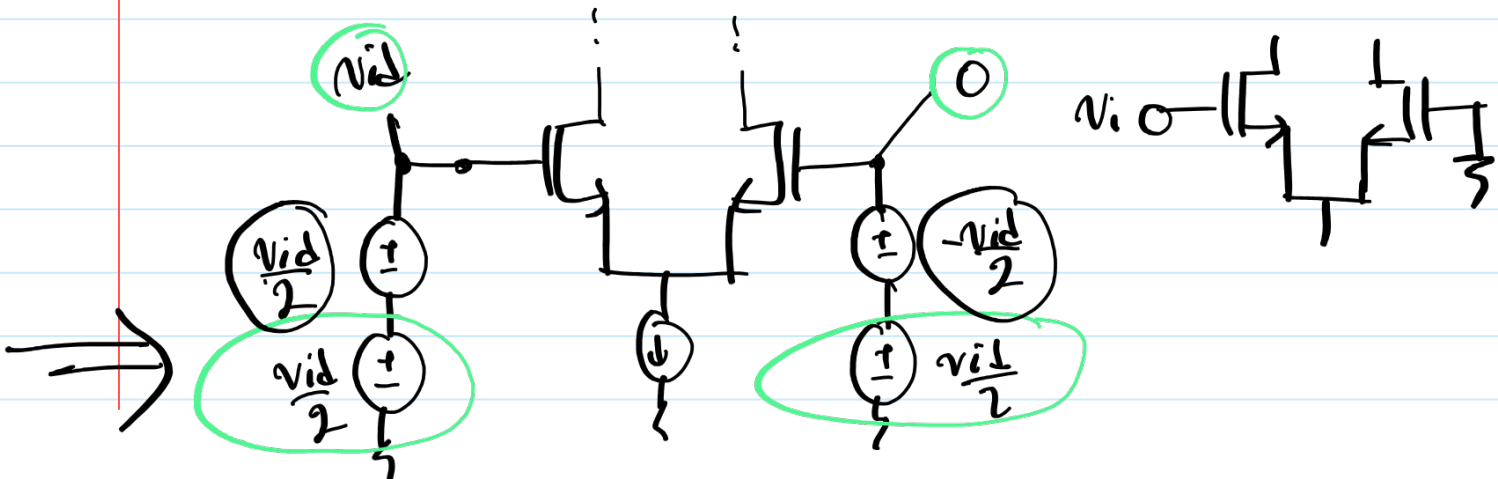


$$-\frac{v_o}{2} = -g_m \frac{v_{id}}{2} R_D$$

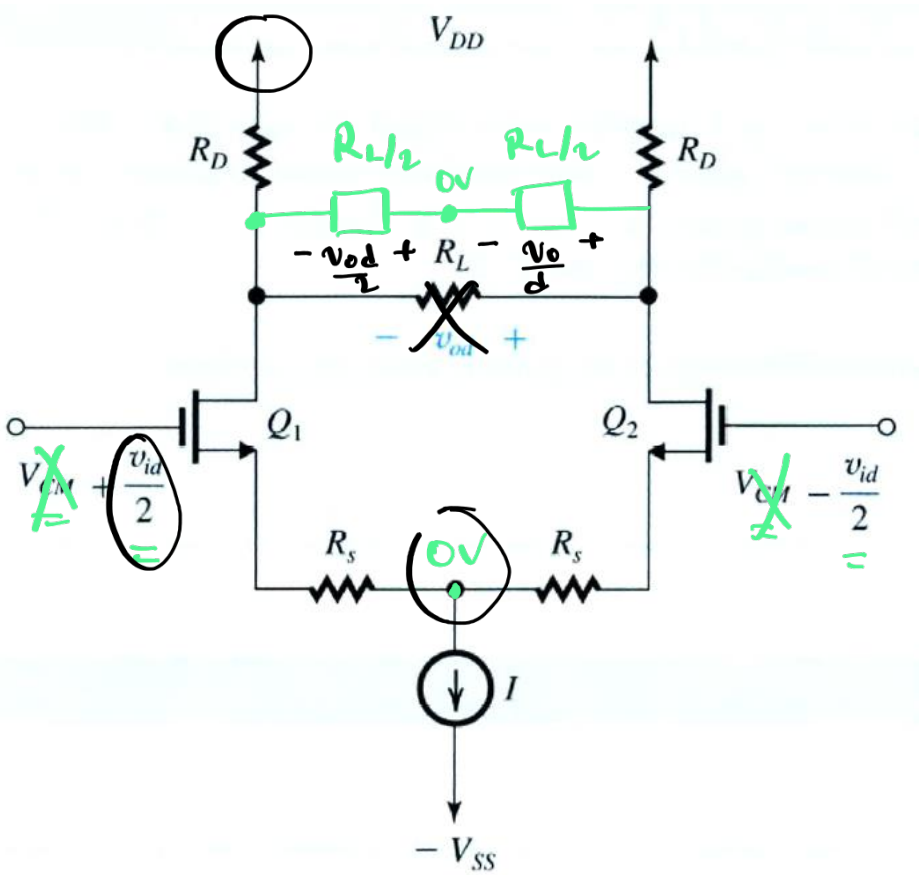
$$A_d = \frac{v_o}{v_i} = g_m R_D$$

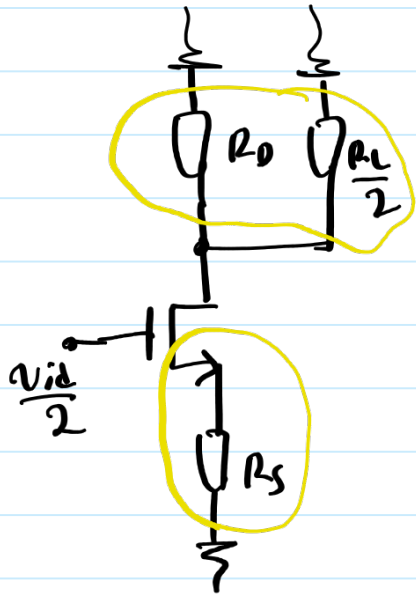


$$A_d = g_m (R_D \| r_o)$$

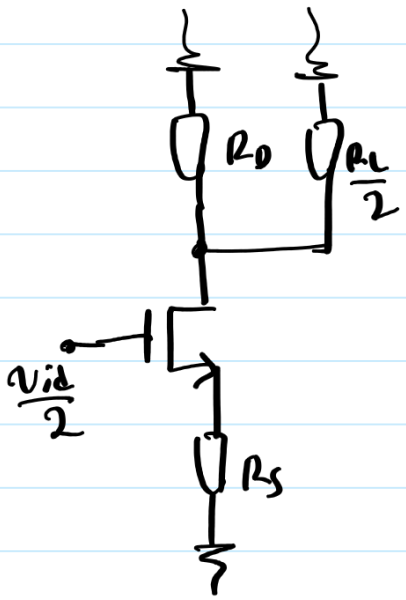


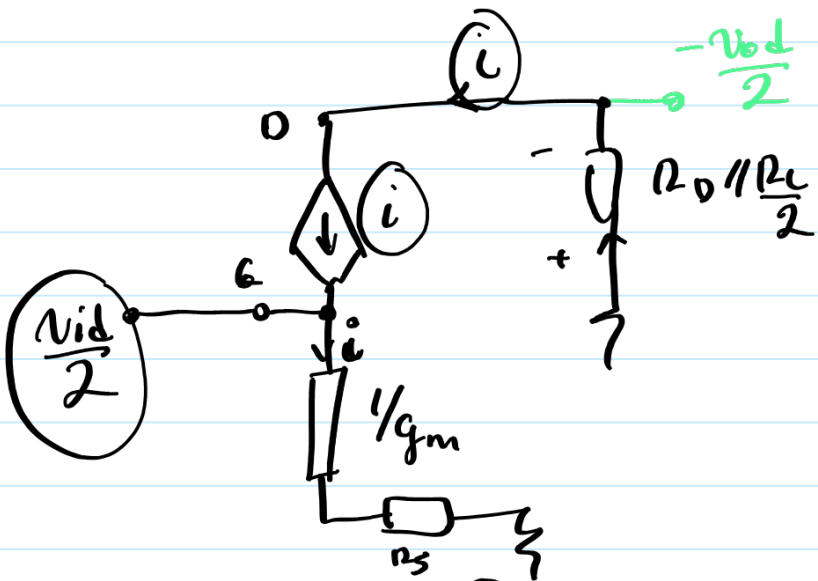
Example





$$A_d \equiv v_{od}/v_{id}$$



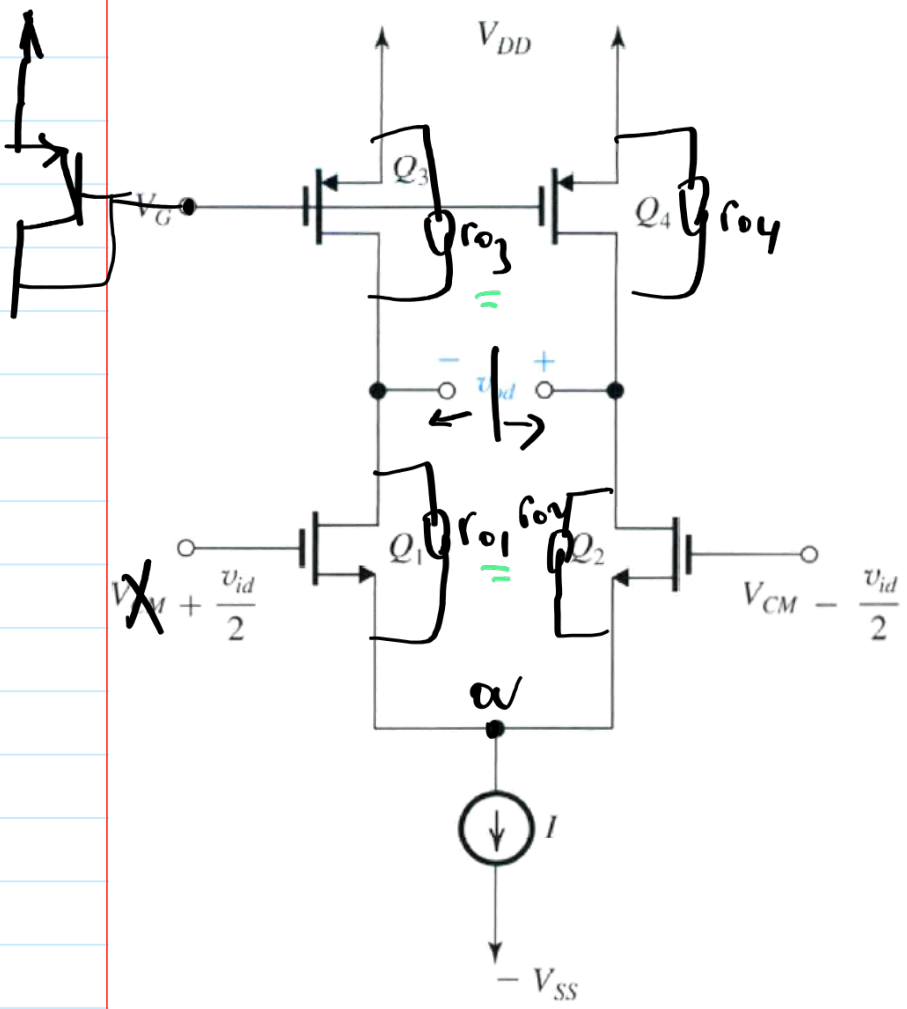


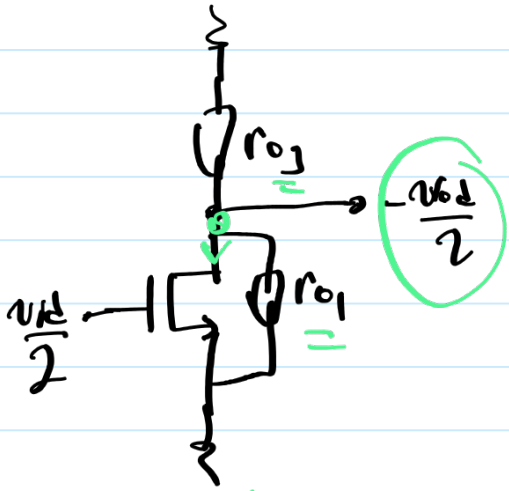
$$\frac{-v_{od}}{2} = -\left(R_D \| \frac{R_L}{2}\right) i$$

$$i = \frac{\frac{v_{id}}{2}}{1/g_m + R_S}$$

$$\frac{v_{od}}{v_{id}} = \frac{(R_D \| R_L/2)}{1/g_m + R_S}$$

The Diff-Amp with Current-Source Load



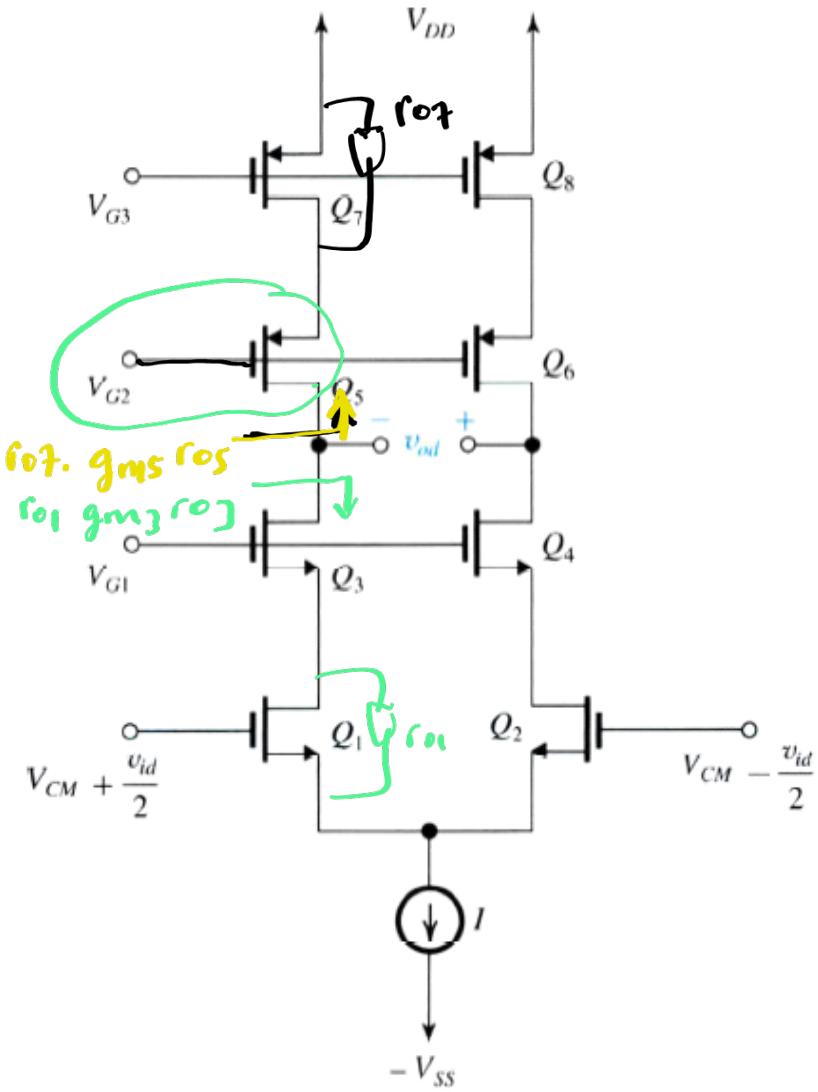


$-g_m \frac{v_{id}}{2} \cdot (r_{o1} \| r_{o3}) =$

$-\frac{v_{od}}{2}$

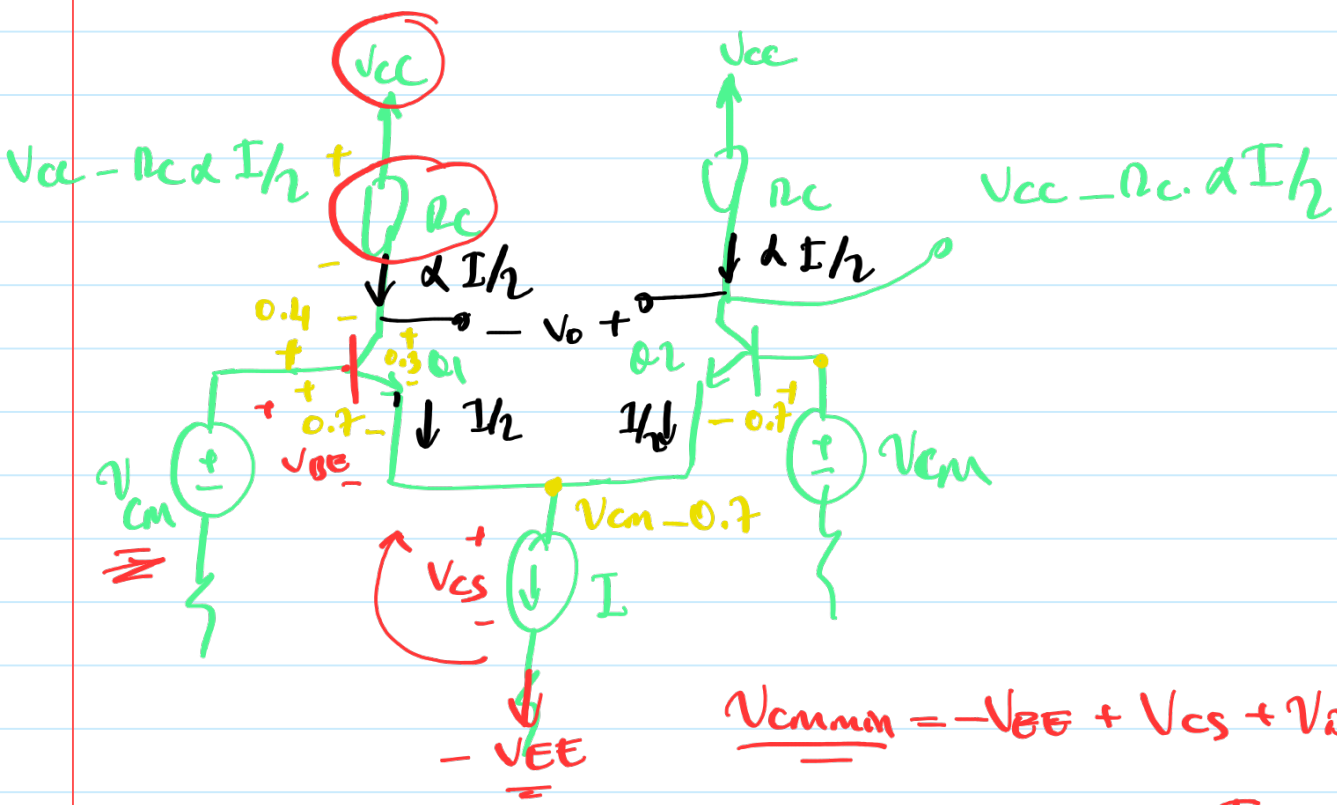
$A_v = \frac{v_{od}}{v_{id}} = g_m (r_{o1} \| r_{o3})$

Cascode Diff-Amp



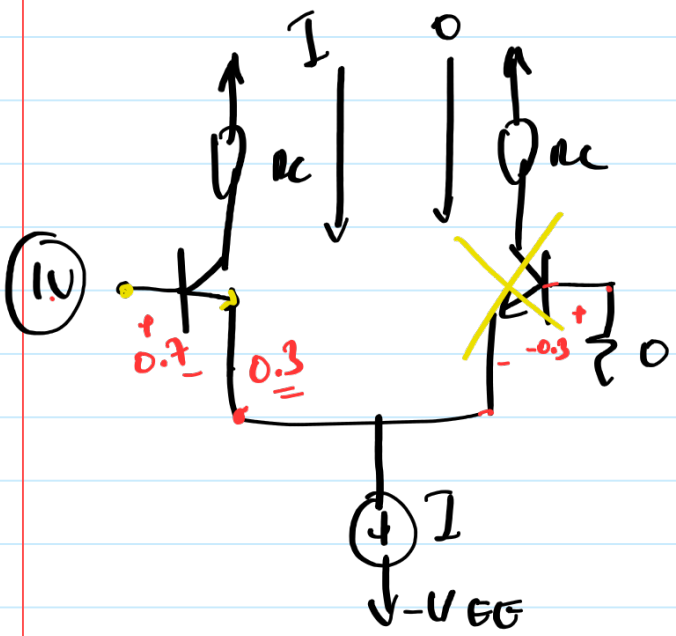
$A_v = g_m(r_{o7} \, g_m r_{o5} \, \| \, r_{o1} g_{m3} r_{o3})$

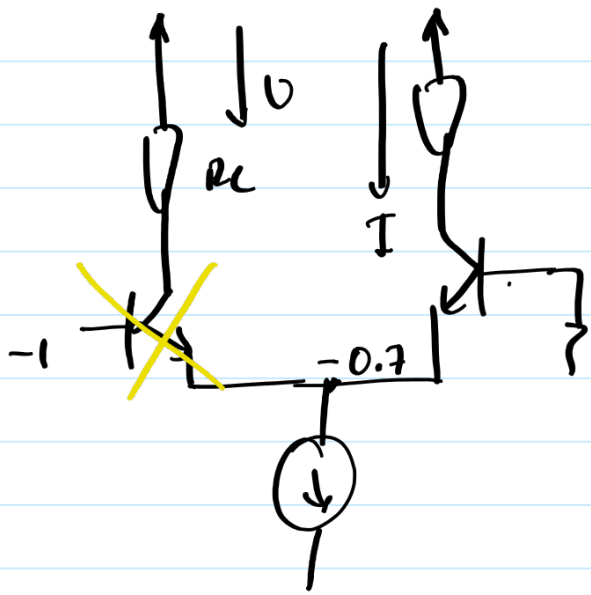
The BJT Differential Pair



$$V_{cm\,min} = -V_{BE} + V_{CS} + V_{BE}$$

$$V_{cm\,max} = V_{CC} - \alpha \frac{I}{2} R_C + 0.4$$

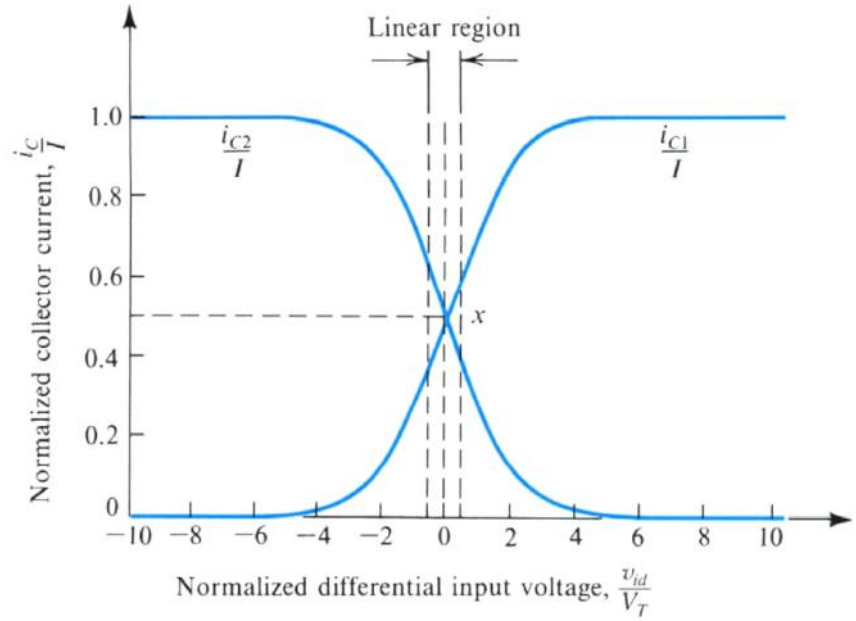


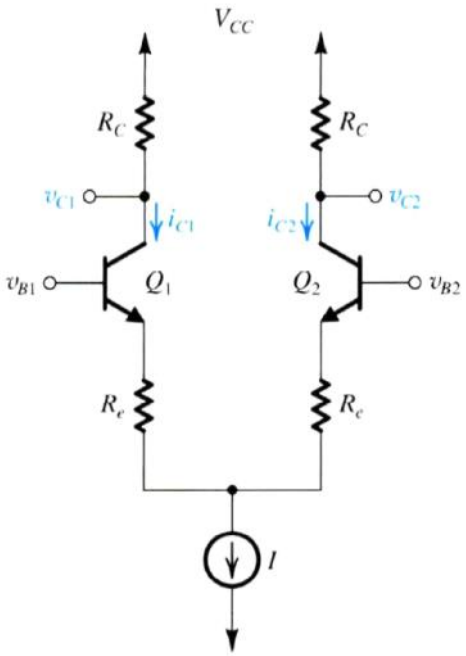


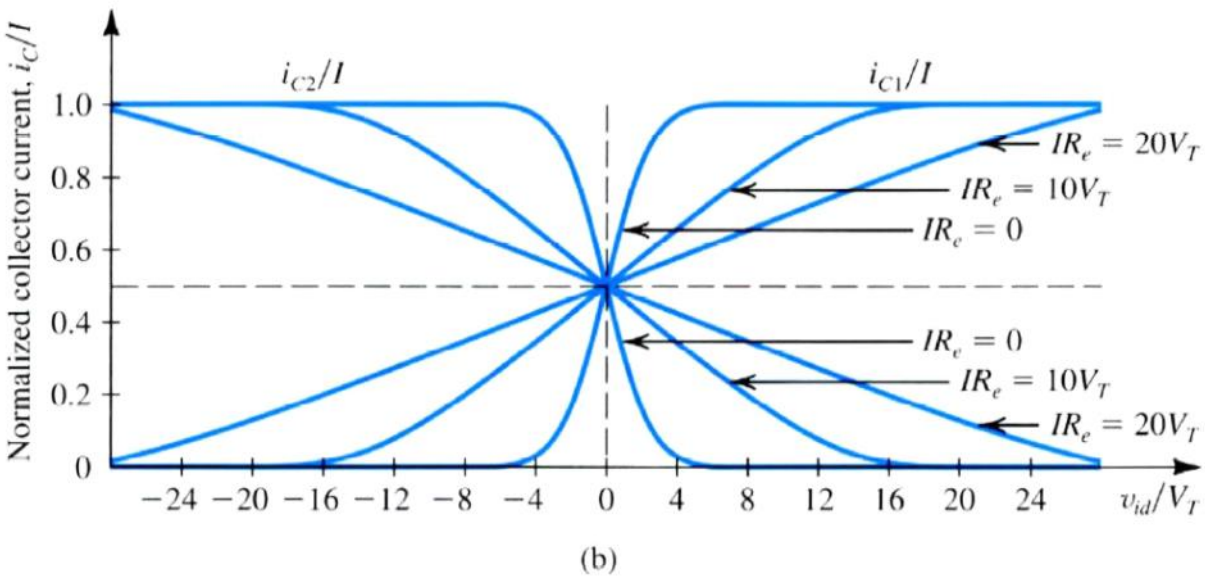
Input Common Mode Range

Large Signal Operation

Large Signal Operation







(b)

Template